Problem Set 494.2 – Multiple Regression Analysis

Submission deadline: The problem set must be submitted to the DV494 Moodle page portal under 'Problem Set Administration' by **midnight Friday Oct 23rd**

The questions are based on material in the lectures, course material, *Mastering Metrics* textbook, and the paper:

* Rodrik, Dani. "Why do more open economies have bigger governments?." *Journal of political economy* 106.5 (1998): 997-1032.

- **This is a compulsory summative problem set that will be assessed Pass/Fail** – please see the Syllabus or 'Problem Set Guidelines' in the DV494 Moodle page for more details.
- Students are allowed (and encouraged) to work on problem sets in groups. Each student must submit their own problem set for assessment, even if it is jointly written.
- Your submitted problem set should not exceed **10 pages** in length, single spaced calibri 11pt font.

  ⇒ ***Hints:*** Please answer all parts directly and succinctly, providing good intuition and/or examples. Statements that are incorrect will be penalized, so be conservatively cautious to ensure you use vocabulary which accurately reflects what you want to say.

  You will find most of the answers *cannot* be directly lifted out of the papers themselves; this is not an exercise that is repeating what you read, but in analytical thinking and critical engagement.

Rodrik (1998)

1. Consider Table 1 from Rodrik (1998) below:

   a. In regression (1) which of the control variables are dummy variables?
      a. Dummy variables are: OECD, Socialist, Latin America, East Asia, Sub-Saharan Africa

   b. Consider the regional control variables Latin America, East Asia, and Sub-Saharan Africa. What regional category(s) are not included in this list? For countries that belong to this excluded regional category, which variable captures ***their*** impact?

The regions that are not part of the regression, are countries which are outside of Latin America, East Asia, and Sub-Saharan Africa. This includes South Asia, the Middle East and North Africa, Western Europe, North America, and the rest of the world. However, these countries constitute the reference group and their baseline effect is being captured by the constant (intercept). Each of the regional dummies' coefficients is the average difference between that region and the reference group, holding all the other variables constant.

c. Hypothetically, if Rodrik had excluded the variables Latin America, East Asia, and Sub-Saharan Africa from regression (1), explain ***in words*** the nature of the '***between'*** variation across regional categories that would be introduced in the estimated relationship between Openness and Government Size.

If Rodrik would have not included those regional controls, this would have introduced between-region variation into the estimated relationship between Openness and Government Size. Consequently, the openness coefficient would have also reflected the differences in average openness and government size across regions, not just across countries within a region. The estimate would have mixed "between-region" and "within-region" variation. If regional characteristics such as history, institutions, culture, and geography, affect both openness and government size, leaving the dummies out creates an omitted variable bias.

d. Since, in fact, Rodrik did include the variables Latin America, East Asia, and Sub-Saharan Africa in regression (1), explain ***in words*** the nature of the '***within'*** regional variation reflected in the estimated country-level relationship between Openness and Government Size.

Since Rodrik does include the regional variables, each region has its own intercept. This essentially absorbs the difference in average government size between regions, and the openness coefficient is then identified only from variation within the regions. It now compares countries to others in the same region and asks whether more open Latin American countries have larger governments than less open Latin American countries, and in the same manner with east-Asia, sub-Saharan Africa, etc. The coefficient of 0.223 in regression 1 says that within a given region, and holding const the other controls, a 1% increase in openness is associated with about a 0.22 percentage points increase in government consumption as a share of GDP. The estimate assumes this slope is the same in every region.

TABLE 1

OPENNESS AND GOVERNMENT SPENDING

	Dependent Variable					
Independent Variable	Log CGAVG9092 (1)	Log CGAVG8589 (2)	Log GIAVG9092 (3)	Log GIAVG8589 (4)	DGOV6092 (5)	DOPEN6092 (6)
Constant	3.289*	3.786*	−1.778***	−4.708	6.426	4.439
	(.536)	(.383)	(.990)	(2.872)	(.936)	(.852)
Log GDP per capita	−.030	−.105***	−.413*	−.013	−.151	−.194
	(.084)	(.063)	(.143)	(.448)	(.133)	(.121)
Log dependency ratio	.642*	.630*	.372	−.304	.387	.146
	(.241)	(.193)	(.499)	(1.457)	(.388)	(.353)
Log urbanization	−.203**	−.136***	−.006	−.556	−.381*	.080
	(.093)	(.075)	(.185)	(.537)	(.123)	(.112)
Socialist	.169	.092	−.559	−1.631***	.924*	.260
	(.130)	(.100)	(.413)	(.909)	(.227)	(.207)
OECD	−.007	−.014	−.051	−.080	.040	.384
	(.144)	(.122)	(.246)	(.851)	(.254)	(.231)
Latin America	−.171	−.218**	−.564**	.122	−.072	−.041
	(.113)	(.094)	(.221)	(.661)	(.191)	(.174)
East Asia	−.206	−.338**	−.193	−.206	−.693	.836
	(.140)	(.130)	(.267)	(.913)	(.228)	(.208)
Sub-Saharan Africa	−.107	−.239**	−.161	.002	−.100	.041
	(.118)	(.101)	(.232)	(.732)	(.194)	(.177)
Log CGAVG6064					−1.308*	−.019
					(.119)	(.108)
Log OPENAVG8089	**.223***		**.534***			
	(.064)		**(.134)**			
Log OPENAVG7584		**.205***		**.835****		
		(.057)		**(.401)**		
Log OPENAVG6064					**.272***	**−.510***
					(.094)	(.086)
Adjusted R^2	.428	.458	.456	.013	.664	.360
Standard error	.317	.313	.558	1.931	.512	.466
Observations	103	125	75	98	99	99

* Significant at the 99 percent level.
** Significant at the 95 percent level.
*** Significant at the 90 percent level.

2. Consider Table 4 from Rodrik (1998) (below):

   a. Which column represents the 'baseline' regression reproduced from earlier in the paper, and what is the main headline result from that baseline regression? What other control variables are included in this baseline regression?
      a. Column (1) is the baseline regression. The headline result is that more open economies have larger governments, as a one percentage point increase in openness is associated with about a 0.3% increase in government consumption as a share of GDP. This is statistically significant at the 95% level ($p<0.05$). The other control variables, are log GDP per capita, log dependency ratio, log urbanization, log urbanization, socialist, OECD, Latine America, East Asia, and Sub-Saharan Africa (as indicated in the table comments below table 4)

   b. In regression (2), what is the primary explanatory variable of interest? Is it statistically significant?
      a. The primary explanatory variable of interest is openness (OPENAVH8089). In regression (2), the coefficient on its own is 0.000 and statistically not significant. However, openness is interacted with export concentration. Its effect on government size is therefore captured jointly by both terms. The interaction is positive (0.011) and significant at the 95% level. Rodrik argues that the significant interaction and insignificant openness term show that openness is associated with large government in countries that are more exposed to external risks (i.e., more concentrated exports).

   c. In regression (2), what general concept is the variable 'CI90' meant to measure?
      a. CI90 refers to the export concentration index, which proxies for exposure to external trade risks. Countries that export only a few sets of commodities are most likely more exposed to external risks, compared to countries that have a more diverse range of exports (Rodrik, 1998, p. 1014).

   d. In regression (2), which variable is an interaction variable? Is the coefficient estimate on the interaction variable positive or negative? Is it statistically significant?
      a. **The OPENAVG X CI90 is the interaction variable. The coefficient estimate is positive (0.011) and statically significant at the 95% confidence interval ($p<0.05$).**

   e. Provide a narrative ***in words***, without resort to any technical terms, explaining the story presented in regression (2).
      a. Countries that trade more tend to have bigger governments, but the link depends on how risky that trade is. For a country that is selling a lot of different products abroad, being more open makes not that much of a difference to the size it its government. Conversely, a country that only sells a few products abroad, being more open goes along

with a bigger government. Essentially, trade exposes a country to shocks from world markets, and the more concentrated its exports are, the bigger these shocks impact the country. Governments in these countries may then grow larger to protect people from that risk, as a sort of insurance against them.

f. How does the fact that the coefficient on OPENAVG8089 in regression (2) is statistically insignificant impact your interpretation from part (e) above?

    a. In the model with the interaction, the coefficient on OPENAVG8089 is the effect of openness only for a country with CI90 = 0, which means perfectly diversified exports. The coefficient being not statistically significant thus means, that for a country with fully diversified exports (hypothetical scenario) openness is not associated with a larger government. This, however, does not mean that openness is not relevant.

g. In regression (3), what general concept is the variable 'TOTDLOGSTD' meant to measure? What is the analytical purpose of regression (3)?

    a. It refers to the terms-of-trade risk (the price risks a country faces on world markets), which is calculated as the standard deviation of changes in the log of the terms of trade. It accounts for international price fluctuations that can impact incomes across the economy of a country. In regression 3 it is being used as second proxy for trade risks, interacted with the openness variable. The idea is to test whether the relationship identified in regression (2) hold with an additional measure. Based on the results it does hold, as the interaction term is positive and statistically significant at the 99% level ($p<0.01$).

h. For students who have traditionally been taught that 'multicollinearity' is a big econometric problem to look out for: Consider that ***both*** CI90 and TOTDLOGSTD are included in regression (4). Is this a problem? More broadly, what do we learn from regression (4)?

    a. Multicollinearity refers to a situation in which two or more of the independent variables are highly correlated. While there is a correlation between the two risk measures, in regression 4 this is not a problem in the sense of biasing the estimates. The potential cost is only that the standard errors are larger, which makes it harder to separate the individual effects of each variable. In regression (4) including both variables is useful as it provides information on which measure is the operative channel. When both interactions are included, the interaction term with TOTDLOGSTD remains statistically significant at the 95% level, while the CI90 interaction term falls to -0.004 becomes insignificant ($p>0.05$). As the TOTDLOGSTD interaction stays signciant despite the high correlation, multicollinearity did not prevent a clear result. Further, it suggests that terms-of-trade volatility is the more relevant source of external risk.

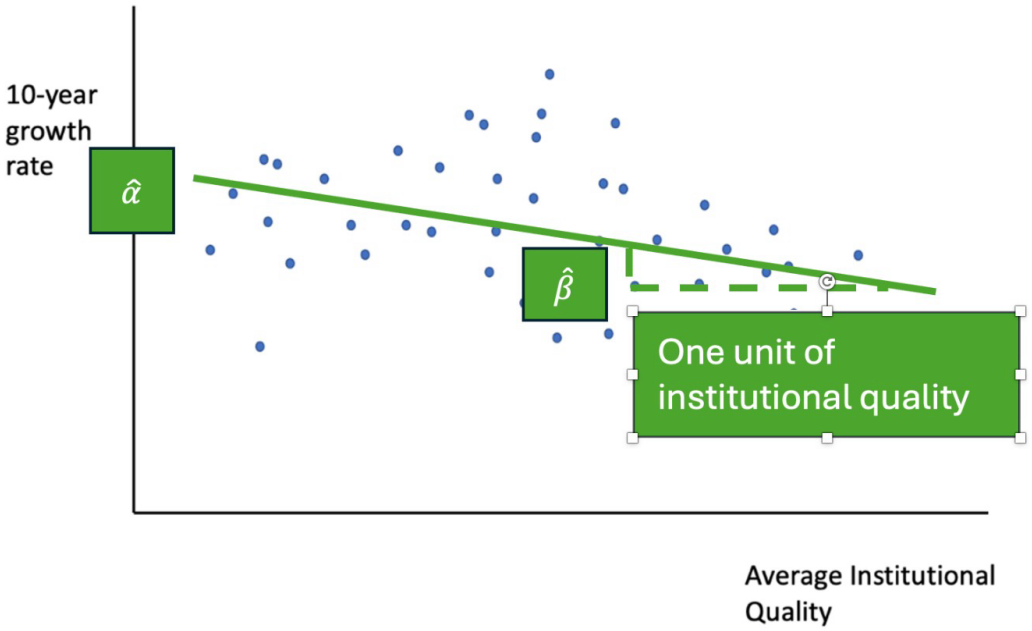
TABLE 4

THE IMPORTANCE OF EXTERNAL RISK

	DEPENDENT VARIABLE: LOG OF REAL GOVERNMENT CONSUMPTION AS A PERCENTAGE OF GDP (Log CGAVG9092)					
INDEPENDENT VARIABLE	(1)	(2)	(3)	(4)	(5)	(6)
OPENAVG8089	.003**	.000	−.003	−.004	−.005	−.004
	(.001)	(.002)	(.002)	(.003)	(.003)	(.003)
CI90		−.661		.429		
		(.366)		(.594)		
OPENAVG8089 × CI90		**.011****		**−.004**		
		(.005)		(.008)		
TOTDLOGSTD			−3.053*	−4.155**	−3.284*	−2.640**
			(1.087)	(1.833)	(1.122)	(1.118)
OPENAVG8089 × TOTDLOGSTD			**.053***	**.064****	**.058***	**.043****
			(.017)	(.027)	(.018)	(.020)
OPENAVG8089 × GDPSH589					2.27E-07	
					(2.68E-07)	
OPENAVG8089 × PRIMSHR90						.003
						(.003)
Observations	103	94	97	92	97	96
Adjusted R^2	.397	.417	.438	.436	.437	.446

NOTE.—Other regressors not shown in the table: constant, log GDPSH585, log DEPEND90, log URBAN90, SOC, OECD, LAAM, ASIAE, and SAFRICA.
* Significant at the 99 percent level.
** Significant at the 95 percent level.
*** Significant at the 90 percent level.

3. $\hat{\alpha}$ Say you are interested in the correlation between growth and institutional quality so you collect data on the average 10-year growth rate and the average level of institutional quality between 2000 and 2010 for a broad sample of countries. You plot the data and find:



a. Just by 'eyeballing' the data, draw a regression line (i.e. 'line of best fit') above. Write out the regression equation for your line and label $\hat{\alpha}$ (the estimated intercept) and $\hat{\beta}$ (the estimated slope coefficient) on the graph above. What is the narrative story your regression line is telling you? [NB no numbers are needed here - just use $\hat{\alpha}$ and $\hat{\beta}$ to denote the intercept and slope, respectively].

   $\widehat{Growth}_i = \hat{\alpha}_o + \hat{\beta} \cdot InstitutionalQuality_i$

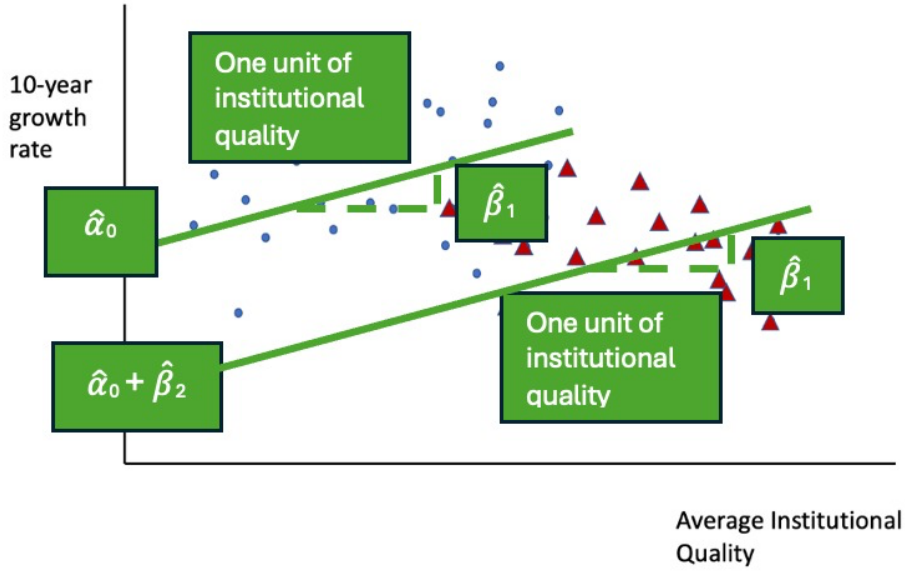
   Narrative story: Countries with higher average institutional quality tend to have lower 10 year growth rate. The plotted relationship is negative, which means that with increasing average institutional quality, the 10year growth rate tends to decrease. However, this is merely correlational and not causational evidence.

b. Now you want to consider the possibility that your analysis may suffer from an omitted variable bias. In particular, you suspect that whether a country is initially a low-income or high-income country might matter, so that leaving this dummy variable out could bias the results. Explain in words why leaving out 'High-Income' from the regression could create an omitted variable bias in this analysis.

   Omiited variable bias arises when the omitted variable is both correlated with the main explanatory variable and affects the outcome. Both conditions could be true in this scenario. High income countries tend to have better institutions, which would be a positive correlation with institutional quality. At the same time, they might grow, in line with the convergence idea, as richer countries have less room for catch-up growth. As high income seems to be associated with better institutions and slower growth, leaving it out makes good institutions look negatively associated with growth, biasing the estimates.

c. You now include the dummy variable 'High Income' (which takes the value 1 if a country is initially a high-income country at the start of the decade, and 0 otherwise) in your regression. Write out the new regression equation.
   a. $\widehat{Growth}_i = \hat{\alpha} + \hat{\beta}_1 \cdot InstitutionalQuality_i + \hat{\beta}_2 \cdot HighIncome_i$.

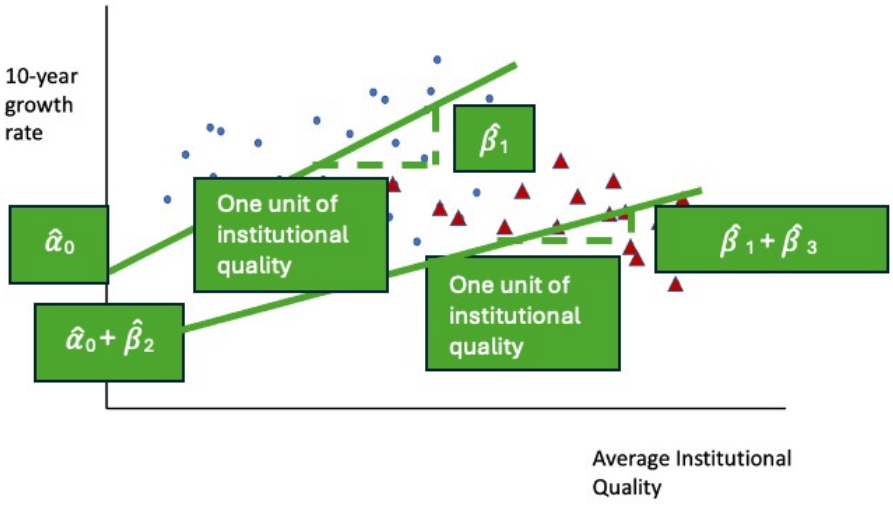
d. The original graph above is replicated below, but now each High-Income country is depicted as a (red) triangle, while lower-income countries are still round dots. Again, 'eyeballing' the data, draw your new regression analysis in the graph, labelling the coefficients from in the regression equation you just wrote out in part (c) on the graph below. What is the narrative story your regression analysis is telling you? How is this different from the story from part (a)?



One we are controlling for initial income, the relationship between institutional quality and growth becomes positive within each income group. This is the reverse of the negative slope evident in (a). The two lines are parallel with common slope. The negative slope in (a) came from comparing two different groups. High-income countries created a misleading downward slope.

e. You have been reading Kremer et al. (2021) and have a theory that differences in Institutional Quality might matter more for growth in low-income countries than in high-income countries. Econometrically, how can you test this theory using your data? Write out your new regression equation below. Again, 'eyeballing' the data, graph your regression results on the graph below, labelling all the coefficient estimates from your written-out regression equation. What do you find? What is the narrative story your regression analysis is telling you?

$\widehat{Growth}_i = \hat{\alpha}_0 + \hat{\beta}_1 \cdot InstitutionalQuality_i + \hat{\beta}_2 \cdot HighIncome_i + \hat{\beta}_3 \cdot (InstitutionalQualility_i \times HighIncome_i)$



Narrative: Institutional quality is associated with faster growth mainly in lower-income countries. In high income countries, growth is low and is less sensitive to differences in institutions. In question (3d) a single common slope was hiding this pattern by averaging a steep positive relationship in poorer countries with a flat one in the richer group. Now that the slope varies based on income group, it shows that the relationship between institutions and growth is clearly different across the two groups. This is in line with the reading of Kremer et al. (2021) and their assumption that differences in Institutional Quality might matter more for growth in low-income countries than in high-income countries.

Compulsory Coding Exercise

This week's exercise focuses on revisiting and 'replicating' Rodrik (1999) using contemporary data. In other words, we will redo part of his analysis using updated data to see if the relationships he found are still significant. To begin, you'll need to download and open the "wdi_forshare" file from the DV494 Moodle page portal under 'Problem Set Administration.' Use your preferred data analysis software. This dataset contains World Development Indicators (WDI) compiled by the World Bank. These are macro-level data, and they provide crucial insights into countries' economic and social conditions.

You can either submit the do-file separately or include your code with your written responses to the questions.

1. Data Preparation:
    a. Familiarize yourself with the dataset by answering these questions. How many countries are recorded? Does the dataset include only countries, or are there regions as well?

    b. The dataset contains more variables than needed. We only need to use "Trade (% of GDP)" and "Expense (% of GDP)" from the 'IndicatorName' column. Keep only these two observations.

c. Since the analysis will be at the country level, remove all region entries from the dataset.

d. The current data format is challenging to use. Convert the wide format to long format using these STATA commands:

reshape long y_, i(CountryCode IndicatorName) j(Year)
reshape wide y_, i(CountryCode Year) j(IndicatorName) string

Tip: Use STATA's "help" function to learn more about reshaping data. Try “help reshape” to understand wide and long formats

e. Rename the variables to remove the “y_” prefix.

f. Determine the number of countries recorded (Hint: use the 'distinct' command (install it with “ssc install distinct”))

g. Analyze missing observations. Tabulate how many observations are recorded for each variable (Hint: use the ‘table’ command)

2. We will use our data to produce a graph comparable to Rodrik Figure 1. As in Rodrik (1998), create a scatter plot with ‘Expense’ on the y-axis, and ‘Trade’ (proxying for “openness”) in the x-axis. Compare your results with Rodrik’s findings. Do you observe a similar pattern?

a. One potential issue is that we are lumping several years into one figure. Create separate scatter plots for the years 2010 and 2019. Briefly explain your observations.

b. There seems to be some outliers and we’d like to label country names on the figures, similar to Rodrik (1998). Label them and discuss why they might be different from the other countries and if dropping these countries from the analysis could be justified.

c. Remove the identified outliers and re-create the scatter plots for 2010 and 2019. Briefly discuss findings.

d. (optional) The World Development Indicators (WDI) dataset includes indicator variables categorized by countries' income levels and regions such as East Asia and Sub-Saharan Africa. How might the patterns differ across these categories? Download the complete dataset from the World Bank website, analyse these categories, and report your findings.